Portfolio Optimizer

User guide: how to use it, which models it uses, and how it makes a plan

For a Robinhood Roth IRA split into an **Investing** and a **Trading** account. The app runs on your own computer, plans each trading day's orders, records what you actually did, and keeps an honest record of how its advice would have performed. **It never places an order**: you place them in the Robinhood app.

Section	What it covers
1. Start and stop	Running the web app
2. The two accounts	Investing, Trading, transfers, cash and settlement
3. A normal day	From opening the app to recording your fills
4. The pages	What each tab and the ticker page show
5. Settings	Risk level, stress measure, growth target, forecasts
6. How a plan is made	From prices to orders, step by step
7. The models	Risk, expected-return, volatility and AI models, and the evidence for each
8. Records, data and costs	Where everything is kept; what is downloaded and paid for
9. Command line	The same things without the browser
10. Limits	What the app cannot know or promise

Plans are estimates from past prices, not promises. A future crash can be deeper than any past one. Nothing here is financial advice.

1. Start and stop

Open PowerShell (or any terminal) and run:

```
cd C:\Users\santo\Desktop\random_projects\Robinhood\TradingAgents
.venv\Scripts\python.exe -m cli.main web
```

The app opens at **http://127.0.0.1:8765/**. Press **Ctrl+C** in that window to stop it. Only one copy can run at a time; if the port is taken, a copy is already running: open the address, or stop it first:

```
Get-NetTCPConnection -LocalPort 8765 -State Listen | ForEach-Object { Stop-Process -Id
$_.OwningProcess }
```

- Use `python.exe -m cli.main`; the `tradingagents` shortcut is broken on this machine.
- `--port 8800` runs it on another port.
- Add `?sleeve=trading` to the address to open the Trading account directly.

2. The two accounts

Robinhood sees **one** Roth IRA. The app divides it into two bookkeeping accounts ("sleeves"); Robinhood's rules (settlement, what an IRA may trade) apply to the whole IRA.

	Investing	Trading
Purpose	The long-term portfolio	An experimental allocation, started with $1,000
Money in	Your recurring contributions ($27.61 a trading day by default)	Only transfers you make from Investing
After a loss	-	Never topped up automatically; its value moves only with its own results and transfers
Settings	Its own risk level, measure, growth target	Its own, separately

Choose the account with the **Investing / Trading** tabs at the top right; each tab shows that account's value. Every page then shows that account's plan. The **Account** card on the Overview shows both and the whole IRA.

Cash, settlement and transfers

- **Settled cash** can be spent now; **unsettled** cash is sale proceeds waiting one trading day (T+1). Buys only ever use settled cash, so there is never a good-faith violation.
- **SGOV** (0-3 month Treasury bills) is money waiting to be invested, not a long-term holding.
- A **transfer** moves shares or settled cash between the accounts. The IRA's total does not change, and the transfer is not counted as profit for either account. It is bookkeeping: nothing is bought or sold at Robinhood.
- If Trading is empty, the Overview offers to fund it from Investing (settled cash first, then SGOV) with a button. It only proposes what Trading still lacks.

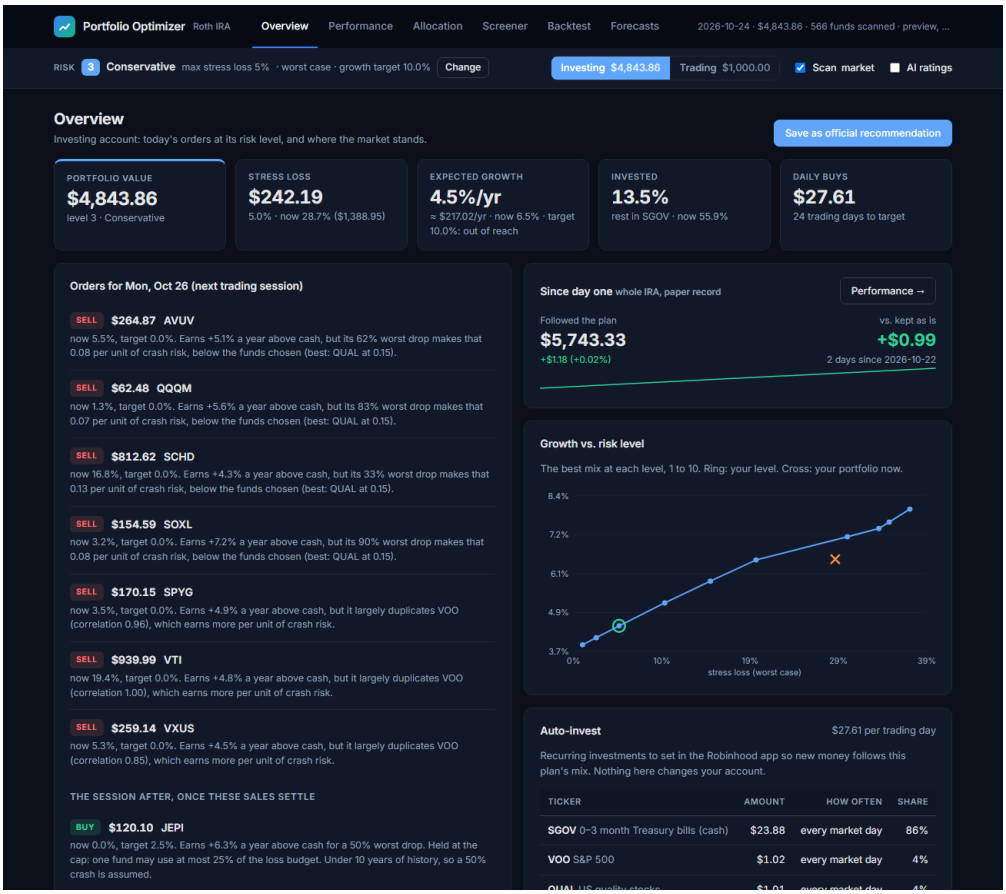
3. A normal day

1. **Open the app** after the market closes (or the next morning). Prices update from the last *completed* trading day; nothing intraday is used.
2. **Pick the account** (Investing or Trading) and read the **Overview**: the orders for the next trading session, what follows once sales settle, and why each one is there.

3. **Save the plan** if you intend to follow it: *Save as official recommendation*. A plan on screen is only a preview; saving makes it the record for its session (see section 8).
4. **Place the orders yourself** in the Robinhood app. The sells come first; the buys use their proceeds the next session, once they settle.
5. **Record what happened**: each fill, or import Robinhood's activity CSV (section 9). Holdings in the app change only through what you record, never because a plan was saved or accepted.
6. **Recurring buys**: if the Auto-invest card lists changes, update your recurring investments in the Robinhood app to match.

Saving or accepting a recommendation is not trading. The app never sends orders.

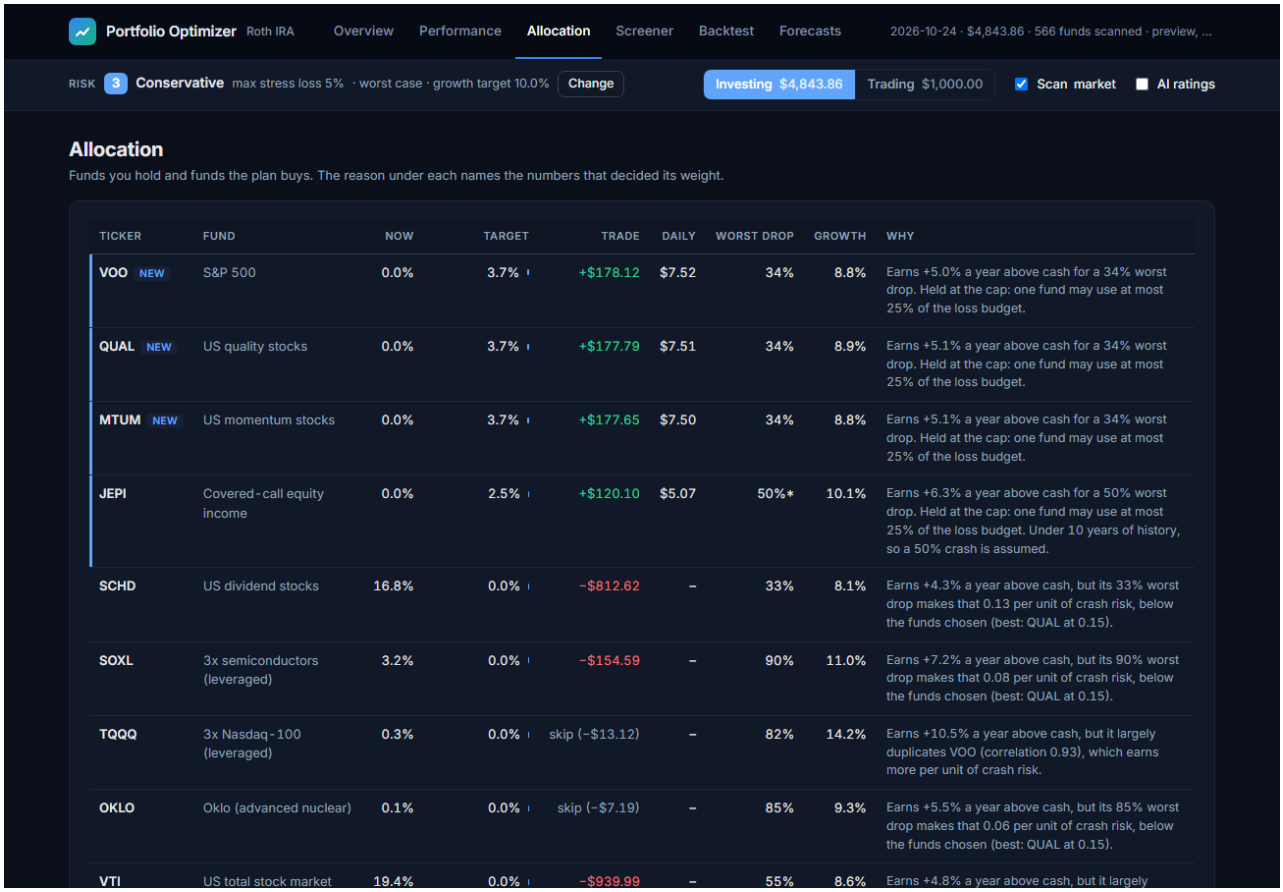
4. The pages



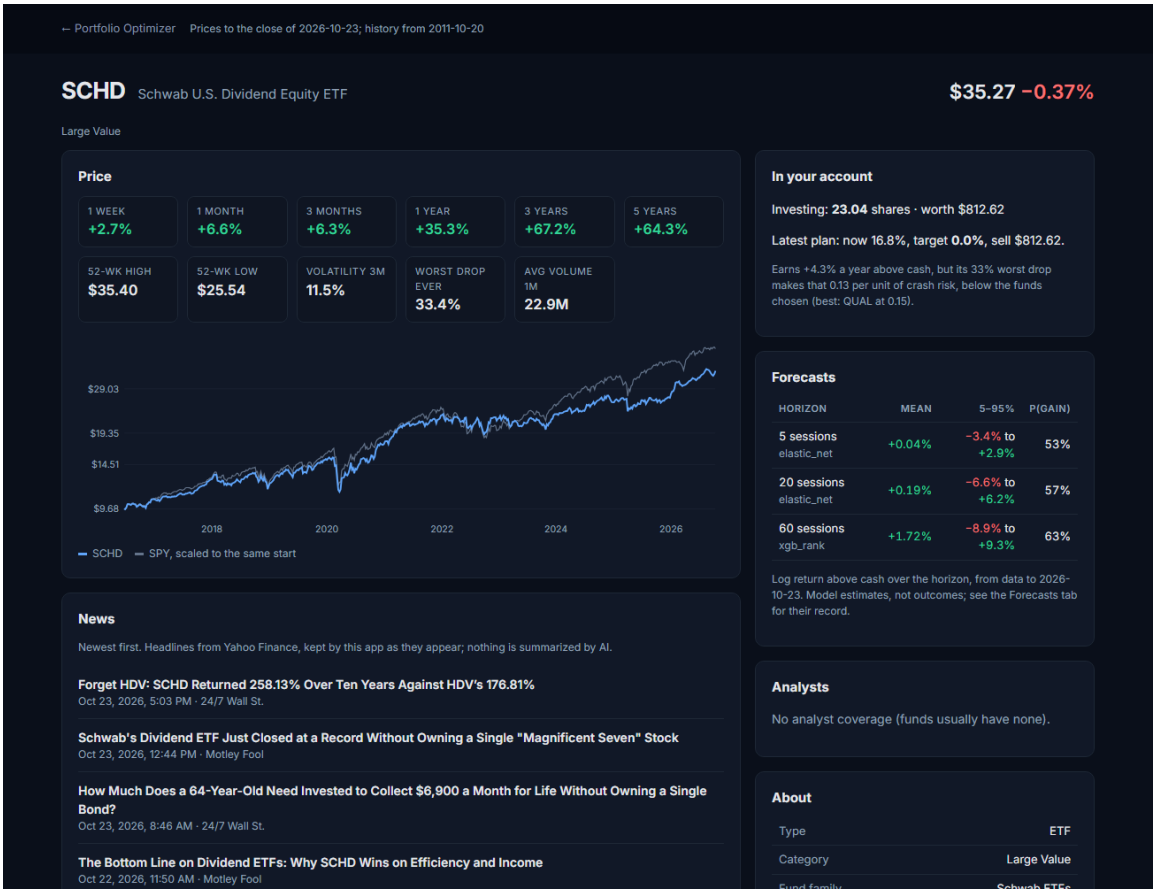
Overview: risk summary and account tabs at the top; orders on the left; paper record, growth chart, auto-invest, account and data freshness on the right.

Page	What it shows
Overview	Value, stress loss, expected growth, share invested, daily buys; the orders for the next session with reasons; the paper record; the growth-vs-risk chart; **Auto-invest** (recurring buys to set in Robinhood and how they differ from yours); the accounts; **Data freshness** (prices, last official recommendation, where expected returns come from).
Performance	The paper record since day one: "followed the plan" against "kept as is", value chart, day by day, and the paper holdings. It covers the whole IRA.
Allocation	Every fund held or chosen: now, target, the order actually placed (small trades under 0.5% of the account are *skipped*), worst drop, expected growth, and why.
Screener	Every fund scanned, ranked by expected growth per unit of crash risk.
Backtest	What the account's value today would have become following the plan month by month since a start date, against holding VTI, SPY or SGOV. A simulation of the past.
Forecasts	Run a forecast model, see each fund's forecast (mean, 5-95% range, chance of a gain), the walk-forward comparison of all models, and the computer's CPU/GPU details.
Settings	Holdings, recurring buys, **Risk and goals** per account, AI ratings, what to scan.

Page	What it shows
Ticker page	Click any ticker. Opens in a new tab: price and chart against SPY, news newest first, analyst price targets and rating changes, earnings surprises, financial statements, your position and the plan's target, forecasts, saved analysis reports.



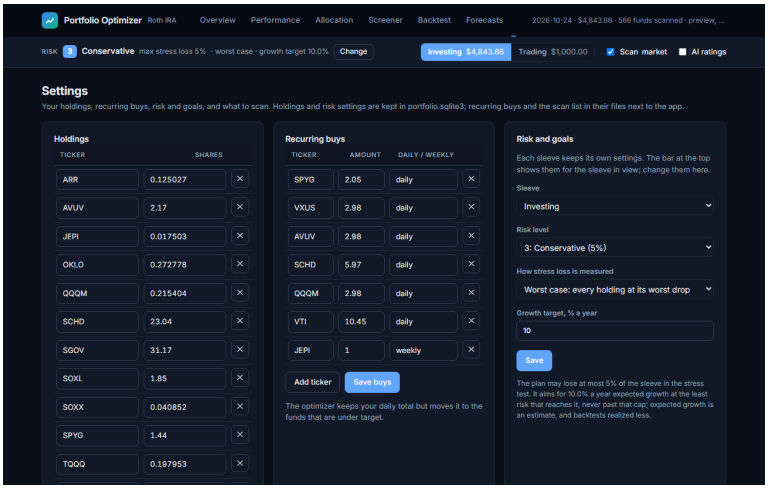
Allocation: the Trade column is the order actually placed; small trades are marked skip.



A ticker page (SCHD): everything about one fund in one place.

5. Settings: risk and goals

Settings > **Risk and goals** holds each account's choices; the bar at the top shows them for the account in view, with a **Change** button that opens this card.



Risk level (1 to 10)

Each level caps the **stress loss**: the most the account may lose in the stress test.

Level	Name	Stress-loss cap	Level	Name	Stress-loss cap
1	Capital preservation	1%	6	Balanced	20%
2	Very conservative	2.5%	7	Growth	30%
3	Conservative	5%	8	Aggressive	40%
4	Cautious	10%	9	Very aggressive	50%
5	Moderate	15%	10	Maximum growth	no cap

How stress loss is measured

- **Worst case** (default, recommended): what the account would lose if every holding repeated its own worst historical drop at the same time. In a 2008-2026 backtest it never broke its cap.
- **Past crashes replayed**: the worst real crash since 1998 (stocks, long bonds, gold, commodities) replayed on the mix, so holdings that did not fall together are not assumed to. It invests more at the same level, but in the same backtest it went past its cap by up to 1.5 times.

Growth target

Optional. With a target, the plan takes the **least** risk expected to reach it, never more than the level allows. If the level cannot reach it, the plan says *out of reach* and stops at the level's cap. Expected growth is an estimate; backtests realized 0.3-1.3 points a year less than expected.

Expected returns: standard rule or forecasts

Forecasts > *Use forecasts in the Investing plan* switches the source of expected returns from the standard rule to a saved forecast model (section 7). In backtests no forecast beat the standard rule after costs, so the standard rule is the recommended setting. The Overview's Data freshness card always says which one is in use.

6. How a plan is made

Every time a page needs a plan, the app runs these steps for the chosen account. The result is reused until prices, your records or your settings change.

Step	What happens
1. Prices	Daily adjusted closes (dividends and splits included) from Yahoo Finance, kept on disk; only sessions not yet stored are downloaded. Only completed sessions are used.
2. Funds to consider	The account's holdings, plus the scan list: about 40 long-lived ETFs, the S&P 500 and "disruptive theme" stocks and funds. A fund needs 3 years of prices. The 60 with the best expected growth per unit of crash risk go on to the optimizer.
3. Risk figures	For each fund: volatility and correlations over 10 years, and its worst drop ever (a young fund borrows an older fund's history, or is assumed to be able to fall 50%).
4. Expected returns	Standard rule: volatility × a Sharpe ratio blended between the fund's own history (at most half) and a common 0.3, less half the variance (what compounding loses). Or a forecast model, if switched on (section 7).
5. Optimize	Choose weights that maximize expected return minus a risk penalty (mean-variance, risk aversion 3), subject to: stress loss within the level's cap; no fund above 35% of the account; no fund using more than a quarter of the cap at its own worst drop; long only. Whatever is not invested stays in SGOV.
6. Growth target	If one is set, the cap is lowered as far as it can go while the plan still expects the target (a bisection over the cap).
7. Orders	Trades under 0.5% of the account are skipped. Sales today; buys the next session from their settled proceeds (T+1). New money comes out of SGOV at your daily contribution pace; an account without contributions (Trading) uses SGOV for the whole gap. Orders under $1 are dropped.
8. Auto-invest	Your daily contribution split across the target mix, at Robinhood's frequencies (every market day, week, 2 weeks or month), with a $1 minimum per order assumed.

The optimizer, in one line

```
maximize expected_return · w - (3/2) · w^T Cov w
subject to 0 <= w <= caps, sum(w) <= 1, stress loss(w) <= cap
```

It is solved by ADMM (the method the OSQP solver uses), and the stress limit and per-fund caps are then enforced exactly. Each reason on the Overview and Allocation pages quotes the numbers that decided a fund's weight.

7. The models

Risk models (what "stress loss" means)

Model	How it works	Evidence (2008-2026 walk-forward)
Worst case (default)	Sum over holdings of weight × that fund's worst historical drop.	Never broke its cap in 274 one-year checks.
Past crashes replayed	Each crash in which SPY, TLT, GLD or DBC fell past a threshold, replayed on the mix; a fund absent from a crash is estimated from its downside beta.	Broke its cap by up to 1.5×; no extra return at equal realized risk.

Expected-return models

The standard rule is used unless you switch to forecasts. The forecast models were all trained and tested the same way: walk-forward from 2012, refitted every year on data known at the time, on the ETF universe.

Model	Type	Inputs
Standard rule (vol_prior)	Volatility × shrunk Sharpe	Each fund's volatility and its own Sharpe history
Momentum	Transparent baseline	12-month return, skipping the last month
Elastic Net	Regularized linear regression	The 22 features below
XGBoost	Gradient-boosted trees (regression, ranking, quantiles)	The 22 features below
LSTM	Small neural network (about 10,000 weights)	Last 60 daily returns of the fund and SPY, plus the features
Chronos-2	Amazon's pretrained forecaster, run locally, zero-shot	Only the fund's last 1,024 daily prices

Features (all as of each date, ranked across funds): momentum over 1, 3, 6 and 12 months; one-week reversal; volatility (all and downside); beta to SPY; distance from the 52-week high and the 200-day average; the past year's deepest fall; high-low range; dollar volume, its trend, and illiquidity. Market conditions: VIX and its slope, the 10-year yield's change, the T-bill rate, SPY's recent return and volatility. For stocks there are also analyst actions (price-target upside, net upgrades) and earnings surprises from Yahoo's dated histories.

What each forecast gives

For each fund and horizon (5, 20 or 60 trading days): the expected log return above cash, the median, a 5-95% range (from the model's errors on data it had not seen), the chance of a gain, and how much the forecast itself depends on which history the model saw (model uncertainty).

What the evidence says

Measure (ETF universe, 2012-2026)	Result
Ranking funds against each other (IC, 0 = chance)	Learned models beat chance and the standard rule: about 0.11 at 20 days and up to 0.21 at 60 days, vs 0.07 and 0.12
Timing the whole market	No model can (correlation about zero or negative)
Chance-of-gain figures	Worse than simply using the historical base rate
Portfolio after trading costs	No forecast beat the standard rule; the best roughly tied it with 5× the trading

Measure (ETF universe, 2012-2026)	Result
Against simply holding VTI or SPY	Every optimizer setting trailed them over 2012-2026 (a strong US bull market)

Volatility and covariance

For volatility, GARCH(1,1) forecasts best at every horizon and EWMA beats the optimizer's 10-year figure. For the optimizer's covariance, the 10-year sample covariance did better than Ledoit-Wolf, factor or EWMA estimates, so it is kept. Full results: docs/FORECASTING.md and docs/BACKTEST_RISK_MODELS.md.

AI analysts (optional, paid)

The TradingAgents multi-agent analysts (by default one, the market analyst) can rate your holdings Buy to Sell, which tilts expected returns. They run only when you press *Run AI ratings*, within a budget ($1 a day and $10 a month by default), and only for models with a configured price. Results are cached; page loads never call them. They are off by default and currently unfunded.

8. Records, data and costs

What	Where	Notes
Holdings, trades, transfers, settings, recommendations, forecast runs	portfolio.sqlite3 (next to my_portfolio.json)	Append-only: a correction is a new record pointing at the old one. Backed up before any upgrade.
Official recommendations	The database, plus a readable report in recommendations/	Saved for a trading session; a later save for the same session supersedes, never edits
Paper record (Performance)	paper_tracker.json	Updated after each close; never restarted by accident
Prices, news, analyst and earnings histories	~/.tradingagents/cache/optimizer/	Downloaded once, then only what is new (news reused for 3 hours, histories for a week)
Forecast models	~/.tradingagents/cache/optimizer/forecast_models/	Each run's model file and its hash

- **Preview vs official.** A plan on screen is not stored. *Save as official recommendation* stores it whole, with the code version, settings and the prices it used, so it can be reproduced later.
- **Which one counts.** A recommendation saved before 3:45 pm New York time is for that day's session; later, for the next session. Its orders cannot fill before the session after its last price.
- **Costs.** Prices, news and data are free (Yahoo Finance, SEC EDGAR, FRED). Only the optional AI analysts cost money, within the budget, and every request is recorded (`book ai`).
- **Computer.** Numerical work runs locally; the GPU (RTX 3050 Ti, 4 GB) is used only where it is measurably faster (large tree models, Chronos-2), one job at a time, falling back to the CPU.

9. Command line

Everything in the web app can also be done from the TradingAgents folder. Prefix each command with `.venv\Scripts\python.exe -m cli.main`.

Command	What it does
web	Start the web app
suggest [--sleeve trading] [--save]	Print the plan and orders; --save makes it official
suggest --risk 4 --model worst_case --target-growth 8	Change the account's settings, then plan
book status	Holdings, settled and unsettled cash, value of each account and the IRA
book fill buy VTI 0.5 330.12	Record a fill you placed (--sleeve trading for Trading)
book contribute 27.61	Record new money deposited into the IRA
book import-csv activity.csv [--commit]	Preview, then import, Robinhood's activity export (duplicates skipped)
book transfer investing trading --ticker SGOV --quantity 2	Move shares between the accounts
book fund-trading [--confirm]	Preview, then make, the transfers that fund Trading

Command	What it does
book recommendations / book decide ID accepted	List official recommendations; record a decision
book ai	AI spending against the budget
track	Update the paper record (the web app also does this)

10. Limits

- **History is not the future.** Stress losses replay the past; a new kind of crash can be deeper.
- **Expected growth runs high.** Backtests realized 0.3-1.3 points a year less than the plan expected.
- **Survivorship.** The S&P 500 list is today's members; backtests use ETFs to limit this bias.
- **Yahoo's analyst and earnings histories** are its lists as of today; corrections cannot be detected.
- **Instruments.** Recommendations cover long stocks and ETFs. Options (level 2 if approved), shorting, futures and crypto are research only in this IRA (docs/CAPABILITIES.md).
- **Sleeves are bookkeeping.** Robinhood sees one IRA; its rules apply to the whole account.
- **No orders are placed.** The app plans and records; you trade.

More detail in the repository: docs/RUNNING.md (commands), docs/FORECASTING.md (models and results), docs/BACKTEST_RISK_MODELS.md (risk models), docs/CAPABILITIES.md (what the IRA allows), docs/AUDIT.md.